fission, as in an atomic bomb). If its nuclei were lighter than iron, energy could be released by joining them together to make iron (nuclear fusion, as in a hydrogen bomb). Once in the form of iron, the matter can release no more nuclear energy by any means whatsoever. The nuclear force holds neutrons and protons together more tightly when they form iron nuclei than when they form any other kind of atomic nucleus.

As the iron is squeezed from its normal density of 7.6 grams per cubic centimeter up toward 100, then 1000 grams per cubic centimeter, the iron resists by the same means as a rock resists compression: The electrons of each atom protest with "claustrophobic" (degeneracy-like) motions against being squeezed by the electrons of adjacent atoms. The resistance at first is huge not because the repulsive forces are especially strong, but rather because the starting pressure, at low density, is very low. (Recall that the resistance is the percentage increase in pressure that accompanies a 1 percent increase in density. When the pressure is low, a strong increase in pressure represents a huge percentage increase and thus a huge "resistance." Later, at higher densities where the pressure has grown large, a strong pressure increase represents a much more modest percentage increase and thus a more modest resistance.)

At first, as the cold matter is compressed, the electrons congregate tightly around their iron nuclei, forming electron clouds made of electron orbitals. (There are actually two electrons, not one, in each orbital—a subtlety overlooked in Chapter 4 but discussed briefly in Box 5.1.) As the compression proceeds, each orbital and its two electrons are gradually confined into a smaller and smaller cell of space; the claustrophobic electrons protest this confinement by becoming more wave-like and developing higher-speed, erratic, claustrophobic motions ("degeneracy motions"; see Chapter 4). When the density has reached $10^5$ (100,000) grams per cubic centimeter, the electrons' degeneracy motions and the degeneracy pressure they produce have become so large that they completely overwhelm the electric forces with which the nuclei pull on the electrons. The electrons no longer congregate around the iron nuclei; they completely ignore the nuclei. The cold, dead matter, which began as a lump of iron, has now become the kind of stuff of which white dwarfs are made, and the equation of state has become the one that Chandrasekhar, Anderson, and Stoner computed in the early 1930s (Figure 4.3): a resistance of ⅗, and